Calculation of $F = D(0,T;0,0)$ by Fourier series

$$F = \int Dy(t)\; e^{\frac{i}{\hbar}S[y]} = \int Dy\; e^{\frac{i}{\hbar}\int_0^T dt\, \frac{1}{2}m(\dot{y}^2-\omega^2 y^2)}$$

where $y(0)=y(T)=0$

Expand $y(t) = \sum_{n=1}^{\infty} a_n \sin\frac{\pi n t}{T}$ (sin function so that BC satisfied)

$$\int_0^T dt\, \dot{y}^2 = \sum_{n,m} a_n a_m \int_0^T dt\, \sin\frac{\pi n t}{T}\sin\frac{\pi m t}{T} = (t = Tx)$$

$$= \sum_{nm} a_n a_m \frac{T}{2}\int_0^1 dx \left\{\cos(\pi n x - \pi m x) - \cos(\pi n x + \pi m x)\right\}$$

$$\int_0^1 dx\, \cos(\pi l x) = \begin{cases} \frac{1}{\pi l}\sin(\pi l x)\Big|_0^1 = 0 & \text{if } l \neq 0 \\ 1, & \text{if } l = 0 \end{cases}$$

$$\Rightarrow \int_0^T dt\, y^2 = \sum_{n=m=1}^{\infty} a_n^2 \frac{1}{2}T$$

$$\int_0^T dt\, \dot{y}^2 = \sum_{n,m} a_n a_m \frac{\pi n}{T}\frac{\pi m}{T}\int_0^T dt\, \cos\frac{\pi n t}{T}\cos\frac{\pi m t}{T}$$

$$= \sum_{n,m} a_n a_m \frac{\pi n}{T}\frac{\pi m}{T}\frac{1}{2}T \underbrace{\int_0^1 dx\, \left(\cos\pi x(n+m) + \cos\pi x(n-m)\right)}_{\delta_{n,m}}$$

$$= \sum_n a_n^2 \left(\frac{\pi n}{T}\right)^2 \frac{1}{2}T$$

Thus $\int_0^T dt\, \frac{m}{2}(\dot{y}^2-\omega^2 y^2) = \frac{1}{4}Tm\sum_{n=1}^{\infty} a_n^2\left[\left(\frac{\pi n}{T}\right)^2 - \omega^2\right]$.

Thus $F = \int Dy \; e^{i \frac{mT}{4\hbar} \sum_{n=1}^{\infty} \left[\left(\frac{\pi n}{T}\right)^2 - \omega^2 \right] a_n^2}$

$$\hookrightarrow \underset{\substack{\uparrow \\ \text{jacobian}}}{J} \, da_1 \, da_2 \ldots da_\ell \ldots = J \prod_{k=1}^{\infty} da_k$$

<u>Note:</u> $Dy(t) \sim \prod_{k=1}^{\infty} da_k$ due to linearity of $y(t) = \sum_{n=1}^{\infty} a_n \sin \frac{\pi n t}{T}$.

Can also calculate the jacobian, but let's try to avoid that:

$$\Rightarrow F = J \prod_{n=1}^{\infty} \int_{-\infty}^{\infty} da_n \; e^{i \frac{mT}{4\hbar} \left[\left(\frac{\pi n}{T}\right)^2 - \omega^2 \right] a_n^2} =$$

$$= J \prod_{n=1}^{\infty} \left(\frac{2\pi i}{\frac{mT}{4\hbar} \left[\left(\frac{\pi n}{T}\right)^2 - \omega^2 \right]} \right)^{1/2} = J \prod_{n=1}^{\infty} \left(\frac{8\pi\hbar i}{mT \left(\frac{\pi n}{T}\right)^2} \right)^{1/2} \cdot$$

$$\cdot \left(\frac{1}{1 - \left(\frac{\cancel{T}\omega \cancel{T}}{\pi n \cancel{T}}\right)^2} \right)^{1/2} = \underset{\substack{\uparrow \\ \text{absorb all } \omega\text{-independent terms in} \\ \text{(still unknown) } \tilde{J}}}{\tilde{J}} \prod_{n=1}^{\infty} \left(1 - \left(\frac{\omega T}{\pi n}\right)^2 \right)^{-1/2}$$

Identity (to be proven in your HW02)

$$\frac{\sin x}{x} = \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{\pi^2 n^2} \right)$$

$$F = \tilde{J}\left(\frac{1}{\prod_{n=1}^{\infty}\left[1-\left(\frac{\omega T}{\pi n}\right)^2\right]}\right)^{1/2} = \tilde{J}\left(\frac{1}{\frac{\sin(\omega T)}{\omega T}}\right)^{1/2} =$$

$$= \tilde{J}\sqrt{\frac{\omega T}{\sin(\omega T)}}$$

Now observe that $F(\omega \to 0) = \tilde{J} = \sqrt{\frac{m}{2\pi i \hbar T}}$

(this fixes unknown $\tilde{J}$, which is ω-independent!)

$\Rightarrow$ finally obtain

$$F = \sqrt{\frac{m}{2\pi i \hbar T}}\sqrt{\frac{\omega T}{\sin(\omega T)}} = \left(\frac{m\omega}{2\pi i \hbar \sin(\omega T)}\right)^{1/2}$$

just as we had before.

Extracting useful info from the propagator

$$D(x',T;x,0) = \sqrt{\frac{m\omega}{2\pi i\hbar \sin(\omega T)}}\; e^{+i\frac{m\omega}{2\hbar}\frac{(x'^2+x^2)\cos\omega T - 2xx'}{\sin\omega T}}$$

Generally, $D(x',T;x,0) = \sum_n \varphi_n(x')\,\varphi_n^*(x)\, e^{-\frac{i}{\hbar}E_n T}$

Consider $J = \int dx\; D(x'=x,T;x,0) = \sum_n \underbrace{\int_{-\infty}^{\infty} dx\, |\varphi_n(x)|^2}_{1}\, e^{-\frac{i}{\hbar}E_n T} =$

$$= \sum_n e^{-iE_nT/\hbar}$$

Let's try that:

$$I = \int_{-\infty}^{\infty} dx\; e^{i\frac{m\omega}{\hbar\sin\omega T}x^2(\cos\omega T - 1)} = \int_{-\infty}^{\infty} dx\; e^{-i\frac{m\omega}{\hbar}\tan\left(\frac{\omega T}{2}\right)x^2}$$

$$\frac{\cos\omega T - 1}{\sin\omega T} = \frac{-2\sin^2\frac{\omega T}{2}}{2\sin\frac{\omega T}{2}\cos\frac{\omega T}{2}} = -\tan\left(\frac{\omega T}{2}\right)$$

$$I = \int_{-\infty}^{\infty} dx\; e^{-\frac{i}{\hbar}ax^2} = \sqrt{\frac{\pi}{ia}}$$

(assuming convergence)

$\mathrm{Re}[+ia] > 0$

$\Rightarrow i(a' + ia'') \Rightarrow$ need $\underline{a'' < 0}$

$$a = \frac{m\omega}{\hbar}\tan\left(\frac{\omega T}{2}\right)$$

Than $I = \sqrt{\frac{\pi\hbar}{im\omega}\,\frac{\cos\frac{\omega T}{2}}{\sin\frac{\omega T}{2}}}$

$$J = \sqrt{\frac{m\omega}{2\pi\hbar i\sin(\omega T)}}\;\sqrt{\frac{\pi\hbar}{im\omega}\,\frac{\cos\frac{\omega T}{2}}{\sin\frac{\omega T}{2}}} = \frac{1}{2i\sin\frac{\omega T}{2}}$$

$$= \frac{1}{e^{i\frac{\omega T}{2}} - e^{-i\frac{\omega T}{2}}} = \frac{1}{e^{i\frac{\omega T}{2}}\left(1 - e^{-i\omega T}\right)} = \sum_{n=0}^{\infty} e^{-i\frac{\omega T}{2}}\, e^{-in\omega T}$$

$$\Rightarrow J = \sum_{n=0}^{\infty} e^{-i\omega T(n+\frac{1}{2})} = \sum_n e^{-iE_nT/\hbar}$$

$$\Rightarrow \omega T\left(n+\frac{1}{2}\right) = \frac{E_nT}{\hbar} \Rightarrow E_n = \hbar\omega\left(n+\frac{1}{2}\right)$$

$n = 0, 1, 2, 3 \ldots$

we have recovered SHO eigenvalues!

Moreover, take a look on $D(x,T;x',0)$ itself:

$$D(x,T;x,0) = \sum_n |\varphi_n(x)|^2 e^{-\frac{i}{\hbar}E_nT} =$$

$$= \sqrt{\frac{m\omega}{\pi\hbar\, 2i\sin(\omega T)}}\, \exp\left(-i\frac{m\omega x^2}{\hbar}\tan\frac{\omega T}{2}\right)$$

$$\tan\frac{\omega T}{2} = \frac{e^{i\frac{\omega T}{2}}(1-e^{-i\omega T})}{i e^{i\omega T}(1+e^{-i\omega T})} = -i\,(1-e^{-i\omega T})\sum_{k=0}^{\infty}(-1)^k e^{-i\omega Tk}$$

Expand in powers of $(e^{-i\omega T})$

$$D(x,T;x,0) = \sqrt{\frac{m\omega}{\pi\hbar}\,\frac{1}{e^{i\omega T}(1-e^{-i2\omega T})}}\; e^{-\frac{m\omega x^2}{\hbar}(1-e^{-i\omega T})(1-e^{-i\omega T}+ e^{-i2\omega T}+\ldots)}$$

$$= \sqrt{\frac{m\omega}{\pi\hbar}}\, e^{-\frac{i\omega T}{2}}\left(1+e^{-i2\omega T}+e^{-i4\omega T}+\ldots\right)^{1/2} \cdot$$

$$\cdot \exp\left\{-\frac{m\omega x^2}{\hbar}\left(1-2e^{-i\omega T}+\ldots\right)\right\} =$$

$$\approx \sqrt{\frac{m\omega}{\pi\hbar}}\, e^{-i\frac{\omega T}{2}}\left(1+e^{-i2\omega T}+\ldots\right)^{1/2} e^{-\frac{m\omega x^2}{\hbar}}\left(1+2\frac{m\omega x^2}{\hbar}e^{-i\omega T}+\ldots\right)$$

$$= \sqrt{\frac{m\omega}{\pi\hbar}}\, e^{-\frac{m\omega x^2}{\hbar}}\left(e^{-i\frac{\omega T}{2}} + \frac{2m\omega x^2}{\hbar}e^{-i\frac{3\omega T}{2}}+\ldots\right)$$

$$= |\varphi_0(x)|^2 e^{-i\frac{E_0T}{\hbar}} + |\varphi_1(x)|^2 e^{-i\frac{E_1T}{\hbar}} + \ldots$$

Thus, using $E_0 = \frac{\hbar\omega}{2} \Rightarrow e^{-i\frac{E_0 T}{\hbar}} = e^{-i\frac{\omega T}{2}}$

we read off $|\varphi_0(x)|^2 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} e^{-\frac{m\omega x^2}{\hbar}}$

$$E_1 = \frac{3\hbar\omega}{2} \Rightarrow e^{-i\frac{E_1 T}{\hbar}} = e^{-i\frac{3\omega T}{2}}$$

$$|\varphi_1(x)|^2 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} e^{-\frac{m\omega x^2}{\hbar}} \frac{2m\omega x^2}{\hbar}$$

$\Rightarrow$ up to a phase: $\varphi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}}$

$$\varphi_1(x) = \sqrt{\frac{2m\omega}{\hbar}}\, x\, \varphi_0(x), \text{ and so on...}$$

* Hence propagator $D(x', T; x, 0)$ does contain a lot of information about quantum system.

Partition function as a path integral

$$Z = \sum_j e^{-\beta E_j} \quad , \quad \beta = \frac{1}{k_B T}$$

$$= \sum_j \langle j | e^{-\beta \hat{H}} | j \rangle$$

$$D(x', T; x, 0) = \langle x' | e^{-i\hat{H}T/\hbar} | x \rangle$$

looks quite similar upon replacement $\frac{T}{\hbar} = -i\beta$

$$D(x', -i\hbar\beta; x, 0) = \langle x' | e^{-i\hat{H}(-i\beta)} | x \rangle =$$

$$= \langle x' | e^{-\beta \hat{H}} \underbrace{\sum_j |j\rangle\langle j|}_{\substack{1 \\ \text{complete set}}} x \rangle = \sum_j e^{-\beta E_j} \langle x' | j \rangle \langle j | x \rangle$$

$$\Rightarrow D(x', -i\hbar\beta; x, 0) = \sum_j e^{-\beta E_j} \langle j | x \rangle \langle x | j \rangle$$

so that $\int dx \; D(x, -i\hbar\beta; x, 0) =$

$$= \sum_j e^{-\beta E_j} \langle j | \underbrace{\int dx \, |x\rangle\langle x|}_{1} j \rangle = \sum_j e^{-\beta E_j}$$

$$\Rightarrow Z = \int dx \; D(x, -i\hbar\beta; x, 0)$$

Check this on SHO

$$D(x',T;x,0) = \sqrt{\frac{m\omega}{2\pi i\,\hbar\,\sin(\omega T)}}\; e^{-\frac{m\omega}{2i\hbar}\frac{(x^2+x'^2)\cos\omega T - 2xx'}{\sin\omega T}}$$

$$\cos\omega(-i\hbar\beta) = \mathrm{ch}(\hbar\beta\omega)\ ,\quad \sin(-i\hbar\beta\omega) = -i\,\mathrm{sh}(\hbar\omega\beta)$$

$$\cos(-ix) = \frac{1}{2}\left(e^{i(-ix)} + e^{-i(-ix)}\right) = \mathrm{ch}(x)$$

$$\sin(-ix) = \frac{1}{2i}\left(e^{x} - e^{-x}\right) = -i\,\mathrm{sh}(x)$$

$$Z = \int_{-\infty}^{\infty} dx \sqrt{\frac{m\omega}{2\pi\hbar\,\mathrm{sh}(\hbar\omega\beta)}}\; e^{-\frac{m\omega}{2\hbar}\cdot 2x^2\cdot\frac{\mathrm{ch}(\hbar\omega\beta)-1}{\mathrm{sh}(\hbar\omega\beta)}}$$

$$= \sqrt{\frac{m\omega}{2\pi\hbar\,\mathrm{sh}(\hbar\omega\beta)}}\sqrt{\frac{2\pi}{\frac{2m\omega}{\hbar}\,\frac{\mathrm{ch}(\hbar\omega\beta)-1}{\mathrm{sh}(\hbar\omega\beta)}}} = \frac{1}{\sqrt{2\left[\mathrm{ch}(\hbar\omega\beta)-1\right]}} =$$

$$\mathrm{ch}\,x - 1 = 2\,\mathrm{sh}^2\frac{x}{2}$$

$$= \frac{1}{\sqrt{4\,\mathrm{sh}^2\frac{\hbar\omega\beta}{2}}} = \frac{1}{e^{\frac{\hbar\omega\beta}{2}} - e^{-\frac{\hbar\omega\beta}{2}}} = \frac{1}{e^{\frac{\hbar\omega\beta}{2}}\left(1-e^{-\hbar\omega\beta}\right)} =$$

$$= e^{-\frac{\hbar\omega\beta}{2}}\sum_{n=0}^{\infty} e^{-\hbar\omega n\beta} = \sum_{n=0}^{\infty} e^{-\hbar\omega(n+\frac{1}{2})\beta} = \sum_n e^{-\beta E_n}$$

$$E_n = \hbar\omega\left(n+\tfrac{1}{2}\right).$$

$$D(x', T; x, 0) = \int Dx(t)\, e^{\frac{i}{\hbar}\int_0^T dt \left(\frac{1}{2} m \dot{x}^2 - V(x)\right)}$$

$$D(x, -i\hbar\beta; x, 0) = \int Dx(t)\, e^{\frac{i}{\hbar}\int_0^{-i\hbar\beta} dt \left(\frac{1}{2} m \dot{x}^2 - V(x)\right)} =$$

$t = -i\tau$, $\tau = \frac{it}{}$ (τ = Imaginary time)

$\frac{dx}{dt} = i\frac{dx}{d\tau}$ $\qquad \tau \in (0, \hbar\beta) = \left(0, \frac{\hbar}{k_B T}\right)$

$$= \int Dx(\tau)\, \exp \frac{i}{\hbar}(-i) \int_0^{\hbar\beta} d\tau \left(i^2 \frac{1}{2} m \left(\frac{dx}{d\tau}\right)^2 - V(x)\right)$$

$$= \int Dx(\tau)\, \exp -\frac{1}{\hbar} \int_0^{\hbar\beta} d\tau \left(\frac{m}{2}\left(\frac{dx}{d\tau}\right)^2 + V(x)\right)$$

$$Z = \int dx\, \underbrace{D(x, -i\hbar\beta; x, 0)}_{\text{density matrix}}$$

(diagonal element of it, more precisely)

- Inverse temperature plays the role of imaginary time

- ~~Transition amplitude for the time Δt is proportional to the classical pa~~